Nadim Mahmud

Prospective MSc student | Robust and Explainable Deep Learning | AI for Agriculture and the Environment

Dhaka, Bangladesh | mahmudnadim479@gmail.com | nadim7009.github.io

ORCID 0009-0002-9274-4850 | linkedin.com/in/nadim-mahmud-ml | github.com/nadim7009

RESEARCH INTERESTS

I study how deep learning models behave when real data stops looking like the benchmark, and how to explain what they learn. I am applying for an MSc to continue this work, with a focus on reliable computer vision for agriculture and the environment.

- Robustness of computer vision models under distribution shift and image corruption (blur, noise, low light), and training methods that close the gap.
- Explainable and trustworthy AI with Grad-CAM, SHAP and probability calibration, to check what a model looks at and whether its confidence holds up.
- Segmentation of microscopy and environmental images with attention-based U-Net models.
- Machine learning for agriculture and the environment: plant disease screening, soil-moisture forecasting, microplastic analysis and carbon-footprint modelling.
- Causal machine learning, especially uplift modelling, to separate what a model predicts from what an action causes.

EDUCATION

Daffodil International University — Dhaka, Bangladesh
B.Sc. in Software Engineering, major in Data Science — *Expected 2026*

- CGPA: 3.73 / 4.00
- Thesis: Mitigating the correlation trap in email marketing through causal uplift modelling

Milestone College — Dhaka, Bangladesh
Higher Secondary Certificate (HSC) — *GPA 5.00 / 5.00*

Asiya Bari Idea School
Secondary School Certificate (SSC) — *GPA 4.79 / 5.00*

PUBLICATIONS

Papers [2] and [3] are accepted at AAI 2026. Paper [1] is under review.

[1] Beyond clean accuracy: controlled robustness evaluation and corruption-aware training for plant disease classification

- **N. Mahmud**, M. S. Hossain, M. H. Riyad, J. J. Bushra, J. J. A. M. Sayor. Submitted to International Journal of Information Technology and Computer Science (IJITCS), MECS Press. Under review.
- My part: the idea, all of the code and the first draft of the manuscript.

[2] Attention-enhanced deep learning for microplastic segmentation in scanning electron microscopy images

- **N. Mahmud** (second author) and co-authors. Accepted at the International Conference on Advances in Artificial Intelligence (AAI 2026), UniMAP, Langkawi, Malaysia, December 2026.
- My part: the original idea and all of the code. Code: github.com/nadim7009/microplastic-sem-segmentation

[3] Explainable machine learning for individual carbon-footprint prediction from traditional, digital and AI behaviours

- **N. Mahmud** (second author) and co-authors. Accepted for oral presentation at the International Conference on Advances in Artificial Intelligence (AAI 2026), UniMAP, Langkawi, Malaysia, December 2026. Code: github.com/nadim7009/explainable-carbon-footprint

RESEARCH EXPERIENCE

Robustness of plant disease classifiers under image corruption — 2026
First author, paper [1]. Idea, all code and first manuscript draft

- Designed a controlled benchmark: Gaussian noise, brightness reduction and Gaussian blur at three severity levels each, applied to the PlantVillage test set (54,305 images, 38 classes).
- Built a group-aware split that keeps all photos of one leaf on the same side, to stop near-duplicate images from inflating validation scores.
- Measured how much accuracy a strong EfficientNetV2-S baseline loses under each corruption and severity.
- Tested corruption-aware training of ResNet50 with each corruption alone and all three together, tracking clean, mean corrupted and worst-case accuracy.
- Compared four architectures and checked model reliability with Grad-CAM, reliability diagrams, expected calibration error and multiclass Brier score.

Attention-gated segmentation of microplastics in SEM images 2026

Second author, paper [2]. Proposed the idea and wrote all of the code

- Trained a baseline U-Net and a four-gate Attention U-Net on 237 labelled SEM images with a fixed group-aware split (167 / 36 / 34).
- Compared the models on Dice, IoU, precision and recall to see where attention gates help and where they cost recall.
- Ran an ablation that removes each attention gate in turn, and robustness tests under noise and reduced brightness and contrast.

Undergraduate thesis: causal uplift modelling 2026

Daffodil International University

- Comparing predictive response models with causal uplift models on the Hillstrom email marketing dataset under injected confounding, to measure how often correlation-based targeting picks the wrong customers.

IrrigaSmart: soil-moisture risk prediction from field sensors 2026

Independent project, github.com/nadim7009/irrigasmart-soil-moisture-risk

- Predicted 12-hour irrigation risk from 31,492 sensor observations on Belgian vegetable farms (KU Leuven open dataset), with a crop-cycle split, calibration analysis and sensor-fault stress tests. HistGradientBoosting reached F1 0.977 on unseen crop cycles against 0.972 for a threshold rule.

WORK EXPERIENCE

Machine Learning Intern Jun 2026 – Sep 2026

FlyRank AI *Remote*

- Prepared and cleaned datasets for machine learning experiments.
- Evaluated models, compared metrics across runs and wrote up the results for the team.

TQ-Tech (startup) Dec 2025 – Apr 2026

Digital Marketing & Business Development Executive Intern (Dec 2025 – Mar 2026); Product Manager & Brand/Business Development Executive (Apr 2026) *Remote*

- As an intern, ran user and market research, including competitor analysis, and supported digital marketing and business development work.
- Promoted to executive in April 2026: managed the company's products and wrote product requirements for the development team.
- Analysed leads, campaign and marketing data and prepared regular performance reports.

TECHNICAL SKILLS

Languages and tools: Python, NumPy, Pandas, Matplotlib, Seaborn, Jupyter, Git, Google Colab, Kaggle

Deep learning: PyTorch, torchvision, CNNs, transfer learning, U-Net, Attention U-Net, ResNet, DenseNet, EfficientNet

Machine learning: scikit-learn, XGBoost, gradient boosting, Random Forest, cross-validation, uplift modelling

Evaluation and explainability: robustness benchmarking, calibration (ECE, Brier score), SHAP, Grad-CAM, permutation importance, ablation studies

CERTIFICATIONS

- Artificial Intelligence Fundamentals, IBM SkillsBuild (2026)
- Introduction to Generative AI, Google Cloud Skills Boost (2026)
- Data Science and Machine Learning with Python for Beginners, Ostad

LEADERSHIP AND SERVICE

Media Executive, DIU Software Engineering Club Oct 2023 – Jan 2026

- Handled media and public relations for club events.

Volunteer

- International Robotics Olympiad; Prothom Alo Math Olympiad; Prothom Alo Bondhushova

REFERENCES

Available on request.